Pradeep kumar Tekmal

Senior Data Engineer

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PROFESSIONAL SUMMARY

Senior Data Engineer with 5+ years of experience designing and building analytics focused data platforms and data products in healthcare and regulated environments. Strong hands-on expertise in **Python**, **SQL**, **Apache Spark**, **Databricks**, **Apache Airflow**, and **dbt** to develop scalable ingestion, transformation, and feature generation pipelines that support analytics and AI use cases. Proven experience connecting complex healthcare datasets, including claims and operational data, to deliver reliable, analytics ready and product ready datasets for internal and external stakeholders. Experienced in building and operating cloud-based data platforms using **AWS S3**, **Redshift**, and modern data warehousing architectures with a strong focus on performance, cost efficiency, and reliability. Demonstrated ability to partner closely with product managers and data scientists, iterate quickly on evolving requirements, and produce clear technical documentation for data architectures and workflows.

CERTIFICATION

AWS Certified : AWS Certified Developer Associate: Credential ID: c076ebe5f5eb475c9ae6a2df23656c8f - Link

TECHNICAL SKILLS

Languages: Python, SQL, Scala, R, Shell Scripting
Big Data: Spark, PySpark, Hadoop, Kafka, Airflow, Databricks, Hive, Presto, Flink
Cloud – Azure: Data Factory, Synapse, Databricks, DevOps, Event Hubs, Blob Storage, Data Lake Gen2
Cloud – AWS: S3, EMR, Glue, Redshift, Lambda, Kinesis, Athena, SageMaker
Data Warehousing: Snowflake, Synapse, Redshift, Teradata, Star/Snowflake Schema, Dimensional Modeling
Databases: SQL Server, Oracle, MySQL, PostgreSQL, MongoDB, Cosmos DB, Delta Lake, Redis
ETL & Integration: SSIS, Informatica, Azure Data Factory, AWS Glue, Talend, CDC
BI & Visualization: Power BI, Tableau, SSRS, Qlik, Looker, Grafana
DevOps: Azure DevOps, Git, Jenkins, Docker, Kubernetes, Terraform, CI/CD
Streaming: Kafka, Spark Streaming, Event Hubs, Kinesis, Real-time Pipelines
Tools: JIRA, Confluence, Visual Studio, PyCharm, Postman, Swagger
Compliance: SOX, PCI-DSS, GDPR, HIPAA, Data Encryption, RBAC, Audit Logging

PROFESSIONAL EXPERIENCE

Bloomberg May 2024 – Present
Senior Data Engineer Chicago, IL

- Built and operated **ingestion** pipelines handling **450B+ market data records per day** across batch and real-time workflows supporting Bloomberg Terminal analytics.
- Developed real-time streaming pipelines using **Kafka** and **Spark Streaming** that ingested data from **300+ global exchanges** delivering low latency risk and portfolio analytics to traders.
- Designed and maintained **Medallion architecture (Bronze, Silver, Gold)** using **Spark** and **SQL**which streamlined high volume market and transaction data for downstream analytics and reporting, improving data consistency.
- Created and maintained **Apache Airflow DAGs** to **orchestrate** end-to-end batch pipelines, handling dependencies, retries, SLA monitoring, and large-scale historical backfills, which increased pipeline reliability.
- Built event-driven ingestion and validation services using AWS Lambda, Python, and Amazon S3, and orchestrated multi-stage workflows aligned with AWS Step Functions patterns, improving failure recovery and reducing pipeline latency by 20%.
- Ran distributed Spark batch processing on **AWS (EMR/EC2)** for datasets exceeding **50 TB per day**, reducing end-to-end processing time by **40%** through optimized transformations.
- Provisioned and managed AWS data infrastructure using Terraform to deploy EMR, S3, IAM roles, and monitoring resources, enabling repeatable and secure environment provisioning
- Designed and maintained scalable data lake architecture using Amazon S3, AWS Glue Catalog, and Amazon Athena, enabling governed, cost-efficient analytics access and reducing data onboarding time by 30%.
- Implemented scalable data pipelines using **Apache Spark**, **Kafka**, and **Python** to ingest, process, and integrate raw data from multiple sources, producing high quality datasets for analytics and machine learning use cases.
- Owned the data transformation and enrichment layer using **dbt**, **SQL**, and **Python**, creating analytics ready and machine learning ready datasets through modular models, incremental processing, and automated validation.
- Tuned Spark workloads using partitioning strategies, caching, and executor sizing, improving query performance by **55%** and reducing compute costs by **$5M annually**.
- Built automated data quality and validation frameworks using **dbt tests**, **Python**, and **Apache Spark**, integrated into **CI CD** pipelines to ensure accuracy, consistency, and reliability of analytical and machine learning datasets.

- Implemented CI/CD pipelines using **Git, Jenkins, and Terraform** to automate build, test, and deployment of Spark, **Airflow, dbt**, and **AWS Lambda** workloads, enforcing peer code reviews and improving release reliability.
- Implemented observability using Amazon CloudWatch metrics, logs, and alarms across EMR, AWS Glue, and AWS Lambda pipelines to monitor latency, failures, and data freshness SLAs, improving production reliability
- Worked in **Agile** delivery using **JIRA**, collaborating closely with **product managers** and **data scientists** and **software engineers** to understand data requirements and iterate data solutions based on feedback, while implementing secure by design access controls using **AWS IAM** and **RBAC**.

CVS HEALTH Apr 2020 – Nov 2022

Data Engineer

- Built and maintained **Azure-based data pipelines** supporting **100M+ members**, integrating data from **10K+ pharmacies** and **9,900+ MinuteClinic's** across claims and clinical systems supporting integrated healthcare services.
- Built scalable **ELT pipelines** with **Databricks**, **Azure Data Factory**, and **PySpark** that ingested **CSV**, **JSON**, and **API based sources** into structured bronze, silver, and gold layers allowing analysts to access fresh data within hours.
- Developed **HIPAA-compliant ETL pipelines** using **Azure Data Factory**, **Databricks**, and **PySpark**, processed over **150M**+ **transactions** per **month** from **EHR and claims** platforms with PHI de-identification.
- Designed enterprise **data warehouse** models in **Snowflake** and **Azure Synapse Analytics** to consolidate pharmacy utilization, clinical analytics, and population health reporting, shortening report generation time.
- Implemented **dbt** based curated marts with standardized transformations, automated tests, and documentation for consistent analytics consumption with incremental loads.
- Designed and supported **enterprise data integration solutions** across **on-premises and Azure environments** using **SSIS**, **Informatica**, **Azure Data Factory**, and **Databricks** which reduced processing time for downstream analytics.
- Delivered **real-time streaming pipelines** using **Event Hubs and Spark Streaming** for prescription and adherence events, enabling care coordination, and medication management which contributes to an 18% reduction in **readmissions**.
- Built automated **medication adherence analytics** for monitoring to identify refill gaps, triggering interventions by improving adherence by 22% and supporting **$18M** in annual cost savings.
- Collaborated with product and analytics teams to translate requirements into **high performance SQL models**, enabling faster report generation and improving data accessibility for business users.
- Created **Power BI dashboards** are used by **8K+ clinicians and pharmacists** to monitor prescription trends and quality metrics.
- Implemented automated **data quality framework**s using **Informatica, Great Expectations** monitoring **400+ pipelines and** reducing **incidents 50%** through automated validation.

Discover Financial Services May 2019 – Mar 2020

Snowflake Data Engineer

- Built transaction pipelines processing **300M+ credit card transactions per month** across **60M+ accounts**, maintaining **99.99% availability** for fraud detection and risk analytics.
- Developed **Snowflake-based dimensional models** supporting investment and regulatory reporting on datasets representing **$7T+ AUM**.
- Implemented end to end **ETL workflows** with **Informatica, SSIS, and AWS Glue** to integrate payment networks, merchant processors, and credit bureau data sources, resulting in faster data availability for analytics.
- Designed and maintained **Apache Airflow DAGs** to orchestrate ETL pipelines, manage dependencies, schedule daily and intraday loads, and support controlled backfills, which improved pipeline reliability and cut manual intervention.
- Designed and implemented **data models across multiple schemas** using **modular, reusable SQL patterns**, data validation, and documentation, which increased data quality and eased onboarding of new data sources.
- Automated **50+ recurring regulatory reports**, reducing manual reporting effort by **70%** while preserving audit accuracy and traceability.
- Implemented **CI/CD pipelines** using **Jenkins and Git** to automate deployment of SQL models, ETL jobs, and **Airflow DAGs**, reducing release cycles by **65%**.
- Supported **real-time trade and transaction analytics** using **Kafka** and **Spark Streaming**, enabling low latency data availability for downstream financial and risk analytics.
- Worked in an **Agile (Scrum) environment**, participating in sprint planning, backlog grooming, daily stand-ups, and retrospectives while collaborating closely with product, risk, and compliance teams.

Education

Lewis University Romeoville, IL

Master of Science in Computer Science *Jan 2023 – Dec 2024*